ALIoT Smart Lab - Full System Status

17 October 2026 · "Design and Development of an Intelligent IoT-Based Smart Research Laboratory Management System" (GIU, Robotics & Automation) · Portal **http://192.168.1.200** (local network **Smart_Lab**, no internet)

In one paragraph. One portal runs on the lab laptop, which also serves as the server. It covers 11 laboratories and 3 roles (student, lab staff, administrator).

- **What it does:** bookings with QR credentials and two-factor door access; live security cameras with person detection; digital twins of two industrial robots; environment, consumables and reusables monitoring; lab PC monitoring and screen sharing; maintenance tickets, alerts and incidents; reports and analytics; n8n automations; two local AI assistants (lab data and project knowledge).
- **Fully equipped so far:** only the Robotics Lab (LAB_01) has the physical devices. The other 10 labs have bookings, pages, images and equipment galleries.
- **State:** the software is deployed and tested. Hardware status is in section 4: much of it is currently off or at the university.

Legend: **working now** · **built, waiting for hardware or approval** · **offline now** · **not built**.

1. The sidebar - what each role sees

Section	Administrator	Lab staff	Student
Operations	System overview · Laboratories	Operations (dashboard) · Laboratories	Dashboard · Laboratories · New booking · My bookings
Access & audit	Bookings · Access & audit · IoT Dashboard · Operations Center · System analytics · Live monitor · Security camera · Reports	Reservations · Access monitor · IoT Dashboard · Operations Center · System analytics · Live monitor · Security camera	-
Facility	Maintenance · System Devices · Lab Equipment · Alerts · Incidents · Notifications	Maintenance · System Devices · Lab Equipment · Alerts · Incidents · Notifications · Users	Support: Robots · My reports · Notifications · Profile
Administration	Users & roles · Settings · Emergency	-	-

Also everywhere: **Report an issue** button, the **portal status** chip (e.g. "Portal OK · 5 devices offline"), **notification bell**, **AI assistant** bubble (bottom right). Pages without a sidebar entry: Welcome, Sign-in/Sign-up, Booking QR, Booking / Session / Asset detail, Robots & Robot detail, Experiments, Simulation (demo), Share (students' view of a shared screen), printable weekly/monthly report.

2. Every page and what it does

Start and personal

Page	What it does
Sign-in / Sign-up	University sign-in; public sign-up creates a student account only. Sign-in throttling against password guessing.
Welcome	Six-section scroll story after sign-in: the lab network, real lab photos, robots (3D models: ABB writer, Kinova, UR5e ...), secure access. Informational only.
Dashboards	**Student:** quick booking, upcoming bookings, finished sessions, free labs now. **Staff:** "what is happening, what needs me, what next" today. **Admin (System overview):** live activity, security events, maintenance, open alerts, equipment, usage charts.

Profile	Account details and lab-access setup (fingerprint / face / card enrolment status, face photo submission).
Notifications	Every notification (portal and n8n), with tabs all / unread / n8n / portal / critical, type filter, search.

Laboratories and bookings

Page	What it does
Laboratories	All 11 labs as cards (photos, category, status, availability), search and category filters.
Lab detail	Hero image, door / controller / camera / face server / fingerprint / RFID / relay status (LAB_01), occupancy, environment panel, usage heatmap, smart bin, safety states, equipment gallery, booking.
New booking	Guided: lab → date → time (real availability, everyone's intervals without names) → purpose → equipment → review → confirmed.
My bookings / Booking detail / Booking QR	Own bookings and cancellation. A booking traced end to end (reservation, credential, both factors, door, exit). The QR page held up to the door camera.
Bookings / Reservations (staff)	Approve or reject booking requests with a reason.

Access, security and monitoring

Page	What it does
Access & audit (Access monitor)	Every door event and session. Filters: granted, denied, expired QR, unknown RFID, wrong lab, biometric failed, identity mismatch. Session detail traces one visit end to end. CSV export of the audit trail.
IoT Dashboard (Command Center)	One wall view of the lab: floor schematic (PC tables, robots, door, camera cone), camera, robot fleet, environment, recent events, incidents, device health. Shows only; sends no commands.
Operations Center	Staff command view fed by one request; refreshes live on door events (no polling).
Live monitor	Full-screen wall display: every lab as a tile (door, lock, who is inside by role, alarms), opens in its own window.
Security camera	All cameras wall · **Live (now both cameras side by side)** with people count, names from the door, identity conflicts · Equipment & door boxes (draw zones) · Events with snapshots and acknowledgement · Recordings (admin, explicit Record) · detection boxes on/off.
Emergency (admin)	The only place the portal can open a door: emergency password, time-limited token in memory only, per-lab hold open, evacuation alarm and message on every portal screen.

Facility

Page	What it does
Maintenance (Issues)	Report an issue in under a minute (what, where, equipment, severity, photos). Ticket list with tabs (new, open, assigned to me, overdue, high priority, resolved). Detail with comments and history. Admin maintenance analytics.
System Devices	Registry of every connected device (master ESP32, ESP32-CAM, face server, sensor nodes, robots, security cameras, PCs), state, last report, per-device keys.

Lab Equipment	**1 · Physical environment:** temperature, humidity, smoke, flame, motion, water, emergency. **2 · Robots & PCs:** • UR5e and FANUC 3D twins with workcell editor, views, replay of recorded movements and reference Home pose. • "Driven from" (pendant / network / hand), and the **Home / STOP master control**. • PC agents: mode, view screen (monitor and quality choice), view all screens, messages, approved apps, lock / log off / restart / shut down. • Share my screen to students. **3 · Consumables:** weighing bins with counts. **4 · Reusables:** NFC smart bin. **Equipment register:** assets with a lifecycle page.
Asset detail	An item's whole life: identity, location, holder, inspections, maintenance due, problems and fixes.
Alerts / Incidents	Alerts list. Incidents: acknowledge, investigate, assign owner, notes, recovery actions, AI hypothesis (labelled), resolve with a reason, reopen, escalation flags, "group similar", evidence by time only.
Robots / Robot detail	Students: availability only. Staff: live twin, every joint (position, velocity, distance from Home), telemetry health, Home calibration, events, master control.

Analytics, reports, administration

Page	What it does
System analytics	• **System health:** every service and device as ONLINE / DEGRADED / STALE / OFFLINE / UNKNOWN / SIMULATED, with last OK, reason and what to check. • **Anomalies:** rules, sensor faults and 7-day baselines. • Every subsystem's history and the freshness of each data source.
Reports	• Usage, access outcomes, arrival delays, traceability charts. • Weekly and monthly management report as a printable PDF. • **Reports Centre:** 7 exports (door, environment, robots, cameras, PCs, consumables, alerts) as CSV or print, each naming its source and warning on stale data. Also the cross-system timeline, inventory discrepancies and equipment lifecycle.
Experiments	Thesis experiments: plan (device, firmware, condition, expected), start / stop; every record during a run is tagged with it (data provenance).
Simulation (demo)	Access-flow simulation for presenting without hardware; never touches the API, database or devices.
Users & roles	Accounts and roles; enrol fingerprint (via the door), face (camera or uploaded photos), RFID card.
Settings	Read-only by design (configuration comes from the server's environment).
AI assistant (bubble)	**Lab data:** exact answers from the database in 30-100 ms (free labs, offline devices, door history, issues, system status, charts). **Project knowledge:** cited passages from the project's documents and code (0.1-0.5 s), and full written answers streamed from the local model in the Project knowledge tab.

3. Behind the pages

Part	What it is
Backend	FastAPI + PostgreSQL in Docker: 27 route modules, 237 paths / 259 operations. Live updates over WebSocket. Every protected route is checked by an automatic test (anonymous callers refused).
Frontend	React + TypeScript + Tailwind, served by nginx on http and https (https needed for browser screen sharing).

Background services	Health monitor (all service checks every 10 s); automatic database backups every 6 h, each checked readable; n8n delivery outbox; robot STOP latch and ownership; screen-share relay (in memory, nothing recorded).
Laptop services	Face server (port 5000); security-camera services for camera 1 and 2 (YOLO26s person detection on the GTX 1650 via DirectML, ports 5100/5101); knowledge service (5199); Ollama (local AI); UR5e and FANUC read-only monitors; PC agent.
Automation (n8n)	• 23 workflow definitions in the repo: booking confirmation and reminders, access-denial intelligence, device-offline escalation, maintenance, daily and weekly reports, sensor thresholds, pre-booking check, post-session, data quality, anomaly detection, hazard escalation, node offline, low stock, unattended stock change, asset overdue, robot offline/fault, robot without booking, PC offline, exam violations, robotics daily, emergency. • Live workflows are never edited from here; which ones are active in the live n8n wasn't checked for this report.
AI models (all local, on the CPU)	Portal assistant: qwen2.5:3b. Knowledge agent: qwen2.5:1.5b with nomic-embed-text search over about 7,600 indexed chunks of project docs, code and notes (incremental rebuild).
Data so far	8 users (1 admin, 2 staff, 5 students) · 11 labs · 23 bookings · 1,309 door events · 24,202 sensor readings · 8,629 robot events · 26 assets · 1 consumables bin · 14 camera zones · ~375 alerts · 4 maintenance tickets · 3,237 audit entries.
Safety design	• Only the Master ESP32 opens the door; the portal only via the admin Emergency page. • Robot control is DRY RUN until your sign-off, and the portal STOP is not an emergency stop. • PC viewing and sharing are visible to the user and audited; there's no covert monitoring and no arbitrary commands. • The production database is never reset; tests use a separate one.

4. Hardware - what exists and its state today

Subsystem	Hardware	State (17 Oct, measured)
Server and network	Lab laptop (Intel i7-9750H 6-core, GTX 1650) at 192.168.1.200 (portal, database, AI, camera analysis). Router **Smart_Lab** (192.168.1.1, local only, 2.4 GHz channel now Auto). Lab tablet UI at .151.	**Laptop and router up**; **tablet not answering**
Door (LAB_01)	Master ESP32 (SmartLab_Master_Portal): PN532 NFC reader, fingerprint sensor, 12 V solenoid lock on a relay (fail-safe wiring fixed 5 Oct), reed switch, buzzer; ESP32-CAM for QR + face (ESP32CAM_Vision_FAST) with the face server on the laptop; door audio amplifier planned.	**Door controller offline since 12 Oct 23:51**; **ESP32-CAM not answering**; **face server online**. Door flow was validated earlier.
Environment (System 2)	ESP32 node: DHT22 temperature/humidity, MQ-2 smoke/gas, 5-channel flame sensor, HC-SR501 PIR motion (water-leak and emergency channels exist in the portal but have no sensor yet).	**Offline since 11 Oct**. Flame detector bench test pending (17% activity, mostly one 3 h block on 9 Oct).
Consumables (System 3)	ESP32 bin: load cell + HX711, 16x2 LCD; count from weight and student declarations.	**Offline since 12 Oct**; needs a staff count for discrepancy checks.
Reusables (System 4)	ESP32 smart bin: NFC tags on each board, NFC reader + LCD, servo lid, load cell (5 kg, calibrated).	**Offline since 12 Oct**.
Robots	**UR5e** (ceiling-mounted, Robotiq 2F-85 gripper) at .102: RTDE read-only monitor; Home/STOP built in DRY RUN. **FANUC ER-4iA** (R-30iB Mate Plus) at .100: position monitor.	**At the university**: no joint data now. UR5e awaits commissioning and sign-off; FANUC interface unknown.
Security cameras	Camera 1: Mercusys MC210 (3 MP) at .155. Camera 2: MC200 at .156 (to be mounted diagonally opposite).	**Camera 1: no picture (at the university)**; **camera 2 live, 7 fps**.

Lab PCs	PC-01 … PC-11 watched as camera boxes; the PC agent is built (monitoring, preview, share, actions).	**Agent verified on the admin laptop only**; student PCs need IT approval.
Power	12 V / 10 A PSU → master switch → WAGO distribution → fused branches → XL4016 bucks to 5 V per node.	-

5. Not built yet (by decision or blocked)

- Remote keyboard/mouse control of PCs (IT decision; use RDP shadowing with consent).
- Live robot motion from the portal (DRY RUN until your sign-off); any FANUC command interface.
- Safety-rated remote emergency stop (needs safety I/O hardware).
- Machine-learning anomaly detection (not enough history yet); vibration, current and energy sensing (no sensors).
- Access-control hardware in labs 02-11 (pages and bookings only).
- Wake-on-LAN for PCs (BIOS/network-card support needed).

6. Upgrade candidates (ideas for planning, not commitments)

Area	Candidate upgrade	Needs
Quick software wins	• Keep the AI search model loaded (first-question delay). • Open the portal on https by default (sharing works everywhere). • Faster PC preview using a long-poll instead of a 1 s poll. • Mobile-friendly alerts (push to phones on the lab Wi-Fi).	Your OK for a knowledge-service restart; nothing else.
Scale to more labs	Door kit (ESP32 + NFC + fingerprint + lock + reed) and an environment node per lab; the portal already models all 11 labs.	Hardware per lab, IT/facilities approval.
Better AI	Run the answer model on a GPU or a second machine (written answers in about 1-2 s instead of 4-8 s); larger model for better wording.	A GPU not shared with the cameras.
Robots	Commission UR5e Home/STOP; FANUC monitoring and control via its licensed interface; approved routines (pick-and-place) with the same safety gates.	Lab session, operator, licences, sign-off.
Predictive maintenance	Current, vibration and energy sensors on robots and machines, then baselines and ML once months of data exist.	Sensors; data over time.
Safety	Hard-wired safety I/O for a real remote E-stop; UPS and power monitoring for the server and router.	Safety hardware; electrician.
Assets	NFC/QR tag on every asset for check-in/out; scheduled maintenance dates for all 26 assets.	Tags; staff time.
Scale and reliability	Dedicated server instead of the laptop; second backup copy off-site; multi-process backend for more simultaneous users.	Hardware; IT.

Sources: the repository (sidebar, pages, routes, firmware, tools, n8n folder), the live database (read-only counts), today's measurements and evidence run. Hardware parts as recorded during bring-up.